Search Go
Climate Outlooks
Climate-Weather Link
El Niño/La Niña
MJO
Teleconnections
AO
NAO
PNA
AAO
Blocking
Storm Tracks
Climate Glossary
Outreach
About Us
Our Mission
Who We Are
Contact Us
Web Team
USA.gov

HOME > Climate & Weather Linkage > El Niño / Southern Oscillation (ENSO) > Relative Oceanic Niño Index (RONI): Historical El Niño / La Niña episodes (1950–Present)

Cold & Warm Episodes by Season ERSSTv6

Relative Oceanic Niño Index (RONI): Historical El Niño / La Niña episodes (1950–Present)

Warm (red) and cold (blue) periods based on a threshold of +/- 0.5°C for the Relative Oceanic Niño Index (RONI), using the 1991–2020 base period [3 month running mean of ERSST.v6 SST anomalies in the Niño 3.4 region (5°N–5°S, 120°–170°W) with average tropical mean (20°N–20°S) SST anomalies subtracted. The difference is then adjusted so the variance equals the original Niño 3.4 index].

For historical purposes, periods of below and above normal SSTs are colored in blue and red when the threshold is met for a minimum of five (5) consecutive overlapping seasons. The RONI is one measure of the El Niño Southern Oscillation, and other indices can confirm whether features consistent with a coupled ocean-atmosphere phenomenon accompanied these periods.

Data availability: 1950–2026

Notice: This page is updated by the 5th of each month. Because of the high frequency filter applied to the ERSSTv6 data (Huang et al., 2025, Part 1 and Part 2, *J. Climate*), RONI values may change up to two months after the initial "real time" value is posted. Therefore, the most recent RONI values should be considered an estimate.

View RONI version 5 | View the Oceanic Niño Index (ONI)

Skip to latest data ↓

Relative Oceanic Niño Index (RONI): The three month running average of the relative Niño 3.4 index

Year	DJF	JFM	FMA	MAM	AMJ	MJJ	JJA	JAS	ASO	SON	OND	NDJ
1950	-1.2	-1.1	-1.0	-1.0	-1.0	-0.7	-0.4	-0.3	-0.3	-0.4	-0.4	-0.5
1951	-0.3	0.1	0.5	0.6	0.6	0.7	0.8	1.0	1.0	1.0	0.9	0.7
1952	0.7	0.3	0.5	0.2	0.0	-0.4	-0.4	-0.1	0.1	0.0	-0.1	0.1
1953	0.4	0.7	0.7	0.7	0.9	0.8	0.7	0.7	0.7	0.8	0.6	0.7
1954	0.6	0.6	0.3	0.0	-0.2	-0.2	-0.4	-0.6	-0.5	-0.3	-0.1	-0.2
1955	-0.1	-0.1	-0.1	-0.4	-0.4	-0.5	-0.4	-0.7	-0.9	-1.3	-1.4	-1.3
1956	-0.8	-0.4	-0.1	-0.1	0.0	0.0	-0.1	-0.2	-0.1	-0.1	-0.1	-0.2
1957	0.0	0.2	0.5	0.8	0.9	0.9	1.0	1.1	1.1	1.2	1.5	1.8
1958	1.9	1.8	1.3	1.0	0.7	0.6	0.5	0.3	0.2	0.2	0.4	0.6
1959	0.8	0.8	0.6	0.5	0.3	0.1	-0.1	-0.2	-0.1	0.0	0.2	0.2
Year	DJF	JFM	FMA	MAM	AMJ	MJJ	JJA	JAS	ASO	SON	OND	NDJ
1960	0.2	0.2	0.2	0.3	0.2	0.1	0.1	0.2	0.3	0.1	-0.1	-0.1
1961	0.0	0.1	0.0	0.1	0.2	0.3	0.2	-0.1	-0.2	-0.2	-0.1	0.0
1962	0.0	0.0	0.0	-0.2	-0.2	-0.1	0.0	-0.1	-0.1	-0.3	-0.3	-0.4
1963	-0.3	-0.1	0.0	0.1	0.0	0.4	0.7	1.1	1.1	1.1	1.1	1.1
1964	1.0	0.6	0.1	-0.4	-0.7	-0.7	-0.6	-0.5	-0.5	-0.6	-0.6	-0.5
1965	-0.2	0.1	0.2	0.4	0.6	1.0	1.3	1.5	1.8	1.9	2.0	1.8
1966	1.5	1.3	1.0	0.6	0.3	0.2	0.3	0.2	0.0	0.0	-0.1	-0.2
1967	-0.3	-0.2	-0.3	-0.3	-0.1	0.2	0.2	0.0	-0.1	-0.1	0.0	0.0

1968	-0.2	-0.4	-0.3	-0.3	0.1	0.3	0.6	0.5	0.4	0.6	0.7	1.0
1969	1.1	1.0	0.7	0.4	0.4	0.3	0.3	0.3	0.6	0.7	0.8	0.7
Year	DJF	JFM	FMA	MAM	AMJ	MJJ	JJA	JAS	ASO	SON	OND	NDJ
1970	0.5	0.2	0.2	0.2	0.1	-0.2	-0.5	-0.6	-0.5	-0.6	-0.5	-0.8
1971	-0.9	-1.0	-0.8	-0.6	-0.5	-0.4	-0.3	-0.2	-0.3	-0.4	-0.4	-0.4
1972	-0.2	0.0	0.3	0.5	0.7	0.9	1.1	1.4	1.6	1.8	2.0	1.9
1973	1.7	1.3	0.6	-0.2	-0.7	-1.0	-1.1	-1.2	-1.3	-1.6	-1.9	-1.9
1974	-1.6	-1.3	-1.0	-0.8	-0.6	-0.5	-0.3	-0.2	-0.2	-0.4	-0.6	-0.3
1975	-0.1	-0.1	-0.3	-0.4	-0.6	-0.7	-0.9	-0.9	-1.0	-0.9	-1.0	-1.1
1976	-1.0	-0.6	-0.3	-0.1	0.0	0.3	0.5	0.7	0.9	1.0	1.1	1.1
1977	1.0	0.9	0.5	0.3	0.3	0.5	0.5	0.5	0.6	0.9	1.1	1.1
1978	1.0	0.7	0.3	-0.1	-0.3	-0.2	-0.3	-0.3	-0.3	-0.1	0.2	0.3
1979	0.3	0.3	0.3	0.2	0.1	0.0	0.0	0.3	0.4	0.5	0.5	0.6
Year	DJF	JFM	FMA	MAM	AMJ	MJJ	JJA	JAS	ASO	SON	OND	NDJ
1980	0.7	0.5	0.3	0.1	0.3	0.3	0.1	-0.1	-0.2	-0.1	0.1	0.1
1981	0.0	-0.2	-0.3	-0.2	-0.3	-0.3	-0.2	-0.2	0.0	-0.2	-0.2	-0.1
1982	0.1	0.2	0.2	0.5	0.7	0.8	0.9	1.1	1.6	2.0	2.2	2.4
1983	2.4	2.2	1.7	1.3	0.9	0.5	0.0	-0.2	-0.6	-0.9	-1.1	-1.0
1984	-0.6	-0.5	-0.5	-0.6	-0.6	-0.5	-0.3	-0.2	-0.3	-0.6	-0.9	-0.9
1985	-0.8	-0.7	-0.7	-0.8	-0.8	-0.5	-0.3	-0.2	-0.2	-0.2	-0.1	-0.2
1986	-0.3	-0.3	-0.2	-0.1	-0.1	0.0	0.3	0.6	0.8	1.0	1.2	1.4
1987	1.5	1.4	1.1	0.9	0.8	0.9	1.1	1.3	1.3	1.2	1.0	0.8
1988	0.5	0.1	-0.4	-0.9	-1.5	-1.7	-1.6	-1.4	-1.4	-1.7	-1.9	-1.9
1989	-1.6	-1.4	-1.1	-0.9	-0.6	-0.4	-0.4	-0.4	-0.3	-0.2	-0.1	0.1
Year	DJF	JFM	FMA	MAM	AMJ	MJJ	JJA	JAS	ASO	SON	OND	NDJ
1990	0.3	0.3	0.3	0.3	0.2	0.2	0.2	0.3	0.3	0.3	0.3	0.4
1991	0.6	0.5	0.4	0.4	0.6	0.7	0.9	0.8	1.0	1.2	1.7	2.0
1992	2.1	2.1	1.9	1.8	1.4	0.9	0.5	0.3	0.1	0.1	0.1	0.4
1993	0.6	0.7	0.9	1.1	1.0	0.8	0.5	0.5	0.5	0.5	0.4	0.4
1994	0.3	0.2	0.2	0.3	0.4	0.4	0.5	0.6	0.8	1.0	1.2	1.3
1995	1.2	0.9	0.6	0.2	-0.1	-0.2	-0.4	-0.6	-0.8	-0.9	-1.0	-1.0
1996	-0.9	-0.7	-0.6	-0.4	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.3	-0.3
1997	-0.2	0.0	0.2	0.5	0.8	1.2	1.5	1.8	2.1	2.2	2.3	2.3
1998	2.1	1.8	1.2	0.8	0.1	-0.5	-1.0	-1.1	-1.2	-1.3	-1.5	-1.6
1999	-1.6	-1.3	-1.0	-0.9	-0.8	-0.8	-0.8	-0.9	-0.9	-1.1	-1.3	-1.6
Year	DJF	JFM	FMA	MAM	AMJ	MJJ	JJA	JAS	ASO	SON	OND	NDJ
2000	-1.5	-1.3	-1.1	-0.9	-0.7	-0.5	-0.3	-0.3	-0.4	-0.5	-0.8	-0.8
2001	-0.8	-0.6	-0.5	-0.4	-0.3	-0.1	0.0	0.0	0.0	-0.2	-0.3	-0.3
2002	-0.2	0.0	0.0	0.1	0.2	0.4	0.6	0.8	1.0	1.2	1.3	1.1
2003	0.9	0.5	0.2	-0.2	-0.3	-0.2	0.0	0.1	0.2	0.2	0.3	0.2
2004	0.2	0.1	0.1	0.1	0.2	0.4	0.5	0.7	0.7	0.7	0.7	0.7
2005	0.5	0.4	0.3	0.3	0.2	0.1	-0.1	-0.1	0.0	-0.2	-0.5	-0.8

2006	-0.9	-0.9	-0.7	-0.5	-0.2	0.0	0.1	0.4	0.6	0.7	0.9	0.9
2007	0.6	0.2	-0.1	-0.3	-0.4	-0.4	-0.5	-0.8	-1.0	-1.3	-1.5	-1.6
2008	-1.7	-1.6	-1.4	-1.0	-0.8	-0.5	-0.2	-0.2	-0.3	-0.4	-0.6	-0.9
2009	-1.0	-0.9	-0.7	-0.4	0.0	0.2	0.4	0.4	0.5	0.9	1.3	1.5
Year	DJF	JFM	FMA	MAM	AMJ	MJJ	JJA	JAS	ASO	SON	OND	NDJ
2010	1.4	1.1	0.7	0.1	-0.5	-1.0	-1.2	-1.4	-1.5	-1.6	-1.6	-1.5
2011	-1.4	-1.2	-1.0	-0.8	-0.5	-0.3	-0.3	-0.4	-0.7	-0.9	-1.0	-1.0
2012	-0.8	-0.6	-0.5	-0.4	-0.3	0.0	0.3	0.4	0.4	0.2	0.0	-0.3
2013	-0.5	-0.6	-0.5	-0.4	-0.5	-0.5	-0.5	-0.4	-0.4	-0.2	-0.2	-0.3
2014	-0.4	-0.4	-0.1	0.1	0.2	0.0	-0.1	-0.1	0.1	0.4	0.5	0.6
2015	0.5	0.5	0.6	0.7	0.9	1.0	1.3	1.5	1.8	2.0	2.2	2.3
2016	2.1	1.8	1.3	0.7	0.1	-0.4	-0.7	-0.8	-0.9	-1.0	-1.0	-0.9
2017	-0.6	-0.4	-0.2	-0.1	0.0	0.0	-0.2	-0.5	-0.6	-0.8	-1.0	-1.1
2018	-1.1	-1.0	-0.9	-0.7	-0.3	0.0	0.1	0.2	0.4	0.6	0.8	0.8
2019	0.7	0.6	0.6	0.5	0.4	0.2	0.0	-0.1	0.0	0.2	0.3	0.2
Year	DJF	JFM	FMA	MAM	AMJ	MJJ	JJA	JAS	ASO	SON	OND	NDJ
2020	0.1	0.1	0.0	-0.3	-0.5	-0.7	-0.8	-0.8	-1.2	-1.4	-1.5	-1.4
2021	-1.3	-1.2	-1.1	-0.9	-0.7	-0.6	-0.6	-0.8	-0.9	-1.1	-1.2	-1.2
2022	-1.2	-1.1	-1.2	-1.3	-1.2	-0.9	-0.9	-0.9	-1.1	-1.1	-1.1	-1.0
2023	-0.9	-0.7	-0.5	-0.3	0.0	0.3	0.5	0.8	1.0	1.3	1.4	1.4
2024	1.1	0.8	0.4	0.0	-0.3	-0.4	-0.5	-0.6	-0.7	-0.8	-0.9	-1.0
2025	-1.1	-0.8	-0.7	-0.5	-0.5	-0.4	-0.4	-0.6	-0.8	-0.9	-1.0	-1.0
2026	-0.9	-0.8	-0.4	0.0	0.5	1.0	1.4	1.7				

Back to top ↑

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